Exposé III. Functoriality of Categories of Sheaves

Complete draft translation

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In Exposé I, 5, the behavior of categories of presheaves with respect to functors between the categories of arguments was studied. In this exposé this study is extended to the case of sites and categories of sheaves (nos. 1 and 2). After introducing the induced topology (no. 3), we turn in no. 4 to the comparison lemma, which will play an important role in Exposé IV. In no. 5, for the patient reader, certain commutative diagrams related to localized categories of the type C/X are studied. In Theorem I, 5.1 smallness hypotheses are imposed on the categories under consideration. These hypotheses are not, in general, satisfied by the sites encountered in practice ($\mathcal{U}$-sites). This forces a few contortions (4.2, 4.3, 4.4).

1 Continuous functors

Definition 1.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *continuous* if, for every sheaf of sets F on C', the presheaf

$$X \longmapsto F(u(X))$$

on C is a sheaf.

This notion of continuity depends a priori on the universe $\mathcal{U}$ for which the two sites are $\mathcal{U}$-sites. Proposition 1.5 shows that it does not depend on it.

1.1.1

Let

$$i : \widetilde{C} \to \widehat{C} \qquad \text{(respectively } i' : \widetilde{C'} \to \widehat{C'}\text{)}$$

denote the canonical inclusion functor from sheaves to presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor

$$u_s : \widetilde{C'} \to \widetilde{C}$$

such that

$$iu_s = \widehat{u}^* i', \qquad \widehat{u}^* F = F \circ u$$

(Exposé I, 5.0).

Proposition 1.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. The following properties are equivalent:

i) The functor u is continuous.

ii) For every object X of C and every sieve $R \hookrightarrow X$ covering X, the morphism

$$u_! R \hookrightarrow u(X)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2 and Exposé I, 5.1).[1]

iii) For every bicovering family $H_i \to K$, $i \in I$, in $\widehat{C}$, the family

$$u_! H_i \to u_! K, \qquad i \in I,$$

is bicovering in $\widehat{C'}$.

iv) There exists a functor

$$u^s : \widetilde{C} \to \widetilde{C'}$$

commuting with inductive limits and "extending u", that is, such that the diagram of canonical functors

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism.

Moreover, when C is no longer necessarily a small site but is a $\mathcal{U}$-site, one still has i) $\Leftrightarrow$ iv), and the functor u^s in iv) is necessarily left adjoint to $u_s : \widetilde{C'} \to \widetilde{C}$, and is therefore determined up to a unique isomorphism.

Proof. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathcal{U}$-site will be given in 4.2. Suppose that C is small. Clearly iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii). Let $H_i \to K$, $i \in I$, be a bicovering family of $\widehat{C}$. For every sheaf F on C', the presheaf $u^* F$ (Exposé I, 5.0) is a sheaf on C. Hence (Exposé II, 5.3) the canonical map

$$\mathrm{Hom}(K, u^* F) \longrightarrow \prod_i \mathrm{Hom}(H_i, u^* F)$$

is bijective. By adjunction (Exposé I, 5.1), it follows that the canonical map

$$\mathrm{Hom}(u_! K, F) \longrightarrow \prod_i \mathrm{Hom}(u_! H_i, F)$$

is bijective. Therefore (Exposé II, 5.3) the family $u_! H_i \to u_! K$, $i \in I$, is bicovering.

ii) $\Rightarrow$ i). Let X be an object of C, let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C'. By Exposé II, 5.3, the map

$$\mathrm{Hom}(u_! X, F) \longrightarrow \mathrm{Hom}(u_! R, F)$$

is bijective. By adjunction (Exposé I, 5.1), the map

$$\mathrm{Hom}(X, u^* F) \longrightarrow \mathrm{Hom}(R, u^* F)$$

[1] "Covering" instead of "bicovering" was not necessarily sufficient; see Example 1.9.3 below.

is bijective. Therefore u^*F is a sheaf.

i) $\Rightarrow$ iv). Use the usual notation $\underline{a}$ and i (respectively $\underline{a}'$ and i') for the associated-sheaf functor and the inclusion into presheaves. For every sheaf G on C, put

$$u^s G = \underline{a}' u_! i G.$$

For every sheaf F on C', one has the sequence of natural isomorphisms

$$\begin{aligned}\mathrm{Hom}(G, u_s F) &\simeq \mathrm{Hom}(iG, iu_s F) \simeq \mathrm{Hom}(iG, u^* i' F) \\ &\simeq \mathrm{Hom}(u_! iG, i'F) \simeq \mathrm{Hom}(\underline{a}' u_! iG, F).\end{aligned}$$

The first isomorphism comes from the full faithfulness of i, the second from the definition of u_s, and the third and fourth from adjunction. Thus u^s is left adjoint to u_s. In particular u^s commutes with inductive limits. For every presheaf K on C and every sheaf F on C', one has natural isomorphisms

$$\mathrm{Hom}(u^s \underline{a} K, F) \simeq \mathrm{Hom}(\underline{a} K, u_s F) \simeq \mathrm{Hom}(K, iu_s F) \simeq \mathrm{Hom}(K, u^* i' F) \simeq \mathrm{Hom}(u_! K, i'F) \simeq \mathrm{Hom}(\underline{a}' u_! K, F),$$

whence a functorial isomorphism

$$u^s \underline{a} K \simeq \underline{a}' u_! K.$$

In particular, using this isomorphism for representable K, and taking Exposé I, 5.4(3) into account, one obtains a diagram, commutative up to isomorphism:

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

iv) $\Rightarrow$ ii). Let $h : C \to \widehat{C}$ (respectively $h' : C' \to \widehat{C'}$) be the functor which sends an object to the presheaf it represents. Consider the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

The upper square is commutative (Remark 1.4(3)), and $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. For every object K of $\widehat{C}$, there is a canonical isomorphism (Exposé I, 3.4)

$$\varinjlim_{(h(X)\to K)\in \mathrm{Ob}(C/K)} h(X) \xrightarrow{\sim} K.$$

Since $u_!$, u^s, $\underline{a}$, and $\underline{a}'$ commute with inductive limits, this gives isomorphisms

$$\varinjlim_{(h(X)\to K)} u^s \underline{a} h(X) \xrightarrow{\sim} u^s \underline{a} K, \qquad \varinjlim_{(h(X)\to K)} \underline{a}' u_! h(X) \xrightarrow{\sim} \underline{a}' u_! K.$$

We have $u_!h = h'u$, and, by hypothesis, an isomorphism $\underline{a}'h'u \simeq u^s\underline{a}h$. Hence there is an isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K,$$

which is immediately checked to be functorial in K. Thus the diagram

$$\begin{tikzcd} \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism. Since $\underline{a}i$ is isomorphic to the identity functor, we have $u^s \simeq \underline{a}'u_!i$, whence the uniqueness of u^s. Let $v : H \to K$ be a bicovering morphism of $\widehat{C}$. Then $\underline{a}(v)$ is an isomorphism (Exposé II, 5.3), and therefore $\underline{a}'u_!(v)$, isomorphic to $u^s\underline{a}(v)$, is an isomorphism. Thus $u_!(v)$ is sent by $\underline{a}'$ to an isomorphism; hence $u_!(v)$ is bicovering (Exposé II, 5.3), which proves ii).

The supplementary assertion, in the case where C is small, has been proved along the way. □

1.2.1

The functor $u^s : \widetilde{C} \to \widetilde{C'}$ of Proposition 1.2(iv) can often be interpreted as an "inverse image" functor for a "morphism of topoi" $\widetilde{C'} \to \widetilde{C}$; cf. Exposé IV, 4.9. Its properties are summarized in the following proposition.

Proposition 1.3. Let $u : C \to C'$ be a continuous functor from a small site C to a $\mathcal{U}$-site C'.

1) The functor u^s is left adjoint to u_s.
2) There is a canonical isomorphism $u^s \simeq \underline{a}'u_!i$.
3) There is a canonical isomorphism $u^s\underline{a} \simeq \underline{a}'u_!$.
4) The functor u^s commutes with inductive limits.
5) When $u_!$ is left exact (cf. Exposé I, 5.4(4)), so is u^s. More generally, u^s commutes with all types of finite projective limits with which $u_!$ commutes.

Proof. Assertions (1)–(4) were proved in the course of the proof of Proposition 1.2. Assertion (5) follows from the isomorphism in (2), using the fact that i commutes with projective limits and that $\underline{a}'$ commutes with finite projective limits (Exposé II, 4.1). □

Remark 1.4. Combining Exposé I, 5.4(3) and Proposition 1.3(3), one obtains a diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r] & \widetilde{C'}, \end{tikzcd}$$

in which the upper square is commutative and the lower square is commutative up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may check that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

1) the composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on sets of objects;

2) the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\underline{a}h"'] & C' \arrow[d, "\underline{a}'h'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes.

More precisely, one can choose $\underline{a}$ and $\underline{a}'$ so as to satisfy condition (1). Once $\underline{a}$ and $\underline{a}'$ have been chosen, one can choose u^s so as to satisfy condition (2).

Proposition 1.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes, let C and C' be two $\mathcal{U}$-sites, and let $u : C \to C'$ be a functor between the underlying categories. Then u is continuous relative to $\mathcal{U}$ if and only if it is continuous relative to $\mathcal{V}$. Moreover, when u is continuous, let $u^s_{\mathcal{U}}$ (respectively $u^s_{\mathcal{V}}$) denote the functor between categories of $\mathcal{U}$-sheaves (respectively $\mathcal{V}$-sheaves) introduced in Proposition 1.2(iv). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u^s_{\mathcal{U}}"] \arrow[d] & \widetilde{C'}_{\mathcal{U}} \arrow[d] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u^s_{\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{1.5.1}$$

where the vertical functors are the canonical inclusion functors, is commutative up to canonical isomorphism.

Proof. We give the proof only in the case where C is $\mathcal{U}$-small. The general case is proved in 4.3. It is clear that if u^* carries every $\mathcal{V}$-sheaf on C' to a $\mathcal{V}$-sheaf on C, then it carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C.

Suppose conversely that u^* carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C. Denote by

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}, \qquad \widehat{C'}_{\mathcal{U}} \hookrightarrow \widehat{C'}_{\mathcal{V}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves, and by $u_{\mathcal{U}!}$ and $u_{\mathcal{V}!}$ the "inverse image" functors for $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves respectively. There is a commutative diagram (cf. the explicit construction of $u_!$ in Exposé I, 5.1)

$$\begin{tikzcd} \widehat{C}_{\mathcal{U}} \arrow[r, "u_{\mathcal{U}!}"] \arrow[d, hook] & \widehat{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widehat{C}_{\mathcal{V}} \arrow[r, "u_{\mathcal{V}!}"] & \widehat{C'}_{\mathcal{V}}. \end{tikzcd} \tag{*}$$

The inclusion functors from $\mathcal{U}$-presheaves to $\mathcal{V}$-presheaves are fully faithful and commute with projective limits. It follows from Exposé II, 5.1(i) that a bicovering morphism of $\mathcal{U}$-presheaves is a bicovering morphism of $\mathcal{V}$-presheaves. Let $R \hookrightarrow X$ be a sieve covering an object X of C. By Proposition 1.2(ii), $u_{\mathcal{U}!}R \to u_{\mathcal{U}!}X$ is bicovering. Using the commutativity of $(*)$ and the preceding observation, it follows that $u_{\mathcal{V}!}R \to u_{\mathcal{V}!}X$ is bicovering. Hence (Proposition 1.2) u^* carries $\mathcal{V}$-sheaves on C' to $\mathcal{V}$-sheaves on C. The commutativity of (1.5.1) follows from the commutativity of $(*)$, from Exposé II, 3.6, and from Proposition 1.3(2). ☐

Proposition 1.6. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. Consider the conditions:

i) u is continuous (cf. Definition 1.1 and Proposition 1.2).

ii) For every covering family $(X_i \to X)_{i \in I}$ of C, the family $(uX_i \to uX)_{i \in I}$ of C' is covering.

There is an implication i) $\Rightarrow$ ii).

Assume that the topology of C can be defined by a pretopology T (Exposé II, 1.3) and that u commutes with fiber products.[2] Then i) and ii) are equivalent, and are equivalent to the following condition:

iii) The functor u carries the covering families of T to covering families of C'.

In particular, if fiber products are representable in C and if u commutes with fiber products, then i) and ii) are equivalent.

Proof. Let us prove i) $\Rightarrow$ ii). Let $(X_i \to X)_{i \in I}$ be a covering family of C. For every sheaf F on C', u^*F is a sheaf. Hence the map

$$\operatorname{Hom}(X, u^*F) \longrightarrow \prod_i \operatorname{Hom}(X_i, u^*F)$$

is injective (Exposé II, 5.1). Thus

$$\operatorname{Hom}(u_!X, F) \longrightarrow \prod_i \operatorname{Hom}(u_!X_i, F)$$

is injective. Hence the family $u_!X_i \to u_!X$, $i \in I$, is covering in C' (Exposé II, 5.1).

[2] In fact, it is enough that u commute with the fiber products occurring in base changes of morphisms coming from covering families for T.

Every covering family of the pretopology T is a covering family of C, whence ii) $\Rightarrow$ iii). Let us prove iii) $\Rightarrow$ i). Let X be an object of C, and let $(X_i \to X)_{i\in I}$ be a covering family in $\mathrm{Cov}(X)$ (Exposé II, 1.3). The morphisms $X_i \to X$ are base-changeable and u commutes with fiber products. Thus the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C'. Then the diagram of sets

$$F(u(X)) \longrightarrow \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (Exposé II, 2.1; Exposé I, 3.5 and 2.12). Hence the diagram

$$u^*F(X) \longrightarrow \prod_i u^*F(X_i) \rightrightarrows \prod_{i,j} u^*F(X_i \times_X X_j)$$

is exact. Thus u^*F is a sheaf (Exposé II, 2.4), and i) follows. The last assertion follows from the preceding argument and from the fact that, when fiber products are representable in C, all covering families of C[3] define a pretopology on C whose associated topology is the topology of C. $\square$

Proposition 1.7. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a continuous functor. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[4] We use the notation of Proposition II, 6.4. The functor u_s commutes with projective limits and therefore defines a functor

$$u_s^\gamma : \widetilde{C}_\gamma \to \widetilde{C'}_\gamma.$$

The functor u_s^γ admits a left adjoint u_γ^s, with the following properties:

1) There is a canonical isomorphism

$$u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma,$$

and u_γ^s commutes with all finite projective limits with which $u_{\gamma!}$ commutes.

2) There is a canonical isomorphism

$$u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}.$$

3) The functor u_γ^s commutes with inductive limits.

4) If u^s is left exact, then the diagram (notation of Exposé II, 6.3.0)

$$\begin{tikzcd} \widetilde{C}_\gamma \arrow[r, "u_\gamma^s"] \arrow[d, "j^\sim"'] & \widetilde{C'}_\gamma \arrow[d, "j'^\sim"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism, and u_γ^s is exact.

[3]Indexed by sets belonging to a suitable universe.

[4]In fact one can show that this condition is always satisfied.

5) Suppose that $j^\sim$ (respectively $j'^\sim$) has a left adjoint $\mathrm{Lib}^\sim$ (respectively $\mathrm{Lib}'^\sim$) (Exposé II, 6.5). There is a canonical isomorphism

$$u^s_\gamma \,\mathrm{Lib}^\sim \xrightarrow{\sim} \mathrm{Lib}'^\sim\, u^s.$$

Moreover, when C is not necessarily a small site but is a $\mathcal{U}$-site, the functor u^γ_s admits a left adjoint u^s_γ. Finally, if $\mathcal{V}$ is a universe containing $\mathcal{U}$, and if $u^s_{\gamma\mathcal{U}}$ (respectively $u^s_{\gamma\mathcal{V}}$) denotes the left adjoint of u_s relative to $\mathcal{U}$ (respectively $\mathcal{V}$), the following diagram, analogous to (1.5.1),

$$\begin{tikzcd} \widetilde{C}_{\gamma\mathcal{U}} \arrow[r, "u^s_{\gamma\mathcal{U}}"] \arrow[d, hook] & \widetilde{C'}_{\gamma\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\gamma\mathcal{V}} \arrow[r, "u^s_{\gamma\mathcal{V}}"] & \widetilde{C'}_{\gamma\mathcal{V}} \end{tikzcd} \tag{1.7.1}$$

commutes up to canonical isomorphism.

Proof. The proof in the case where C is a small site is left to the reader. It will be completed in 4.3 in the case where C is a $\mathcal{U}$-site. □

Notation 1.8. From now on we shall use the notation u_s for the functor u^γ_s. This notation causes no confusion, because the functor u_s defined on sheaves of sets commutes with finite projective limits.

Similarly, when u^s is left exact, we shall use the notation u^s for the functor u^s_γ. This abuse of notation is justified by Proposition 1.7(4).

Example 1.9.1. Let X be a small topological space. Let $\mathrm{Open}(X)$ denote the category of open subsets of X, whose objects are the open subsets of X and whose morphisms are inclusions, endowed with the canonical topology. A family $(U_i \subset U)_{i\in I}$ is covering if and only if the open subsets U_i cover U (Exposé II, 2.6). Thus sheaves of sets on $\mathrm{Open}(X)$ are sheaves of sets in the sense of Godement [TF]. Let $f : X \to Y$ be a continuous map. It gives a functor

$$f^{-1} : \mathrm{Open}(Y) \to \mathrm{Open}(X),$$

which sends an open set U of Y to its inverse image $f^{-1}(U)$. The functor f^{-1} is continuous (Proposition 1.6). The functor

$$f^{-1}_s : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$$

is the direct image functor for sheaves in the sense of [TF]. The functor

$$(f^{-1})^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

is the inverse image functor for sheaves in the sense of [TF]. The functor $(f^{-1})^s$ commutes with finite projective limits, by Proposition 1.3(5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subspace. Every open subset of Y is an open subset of X, hence there is a functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X).$$

The functor Φ is continuous (Proposition 1.6). The functor $\Phi_s : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$ is the "restriction to Y" functor. The functor

$$\Phi^s_{\mathbf{Ab}} : \widetilde{\mathrm{Open}(Y)}_{\mathbf{Ab}} \to \widetilde{\mathrm{Open}(X)}_{\mathbf{Ab}}$$

(Exposé II, 6.3.3; Exposé I, 1.7) is the "extension by zero outside Y" functor introduced in [TF]. The functor

$$\Phi^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

(Proposition 1.3) is analogous to extension by zero: for every sheaf H on Y and every point $y \in Y$, the stalk of $\Phi^s H$ at y is canonically isomorphic to the stalk of H at y; for every point $x \in X$ with $x \notin Y$, the stalk of $\Phi^s H$ at x is empty. The functor Φ^s is called the "extension by the empty set outside Y" functor. It commutes with fiber products and with finite products over a nonempty set of indices, but it does not carry the final object of $\widetilde{\mathrm{Open}(Y)}$ to the final object of $\widetilde{\mathrm{Open}(X)}$.

Example 1.9.3. Let C and C' be two small sites, and let $u : C \to C'$ be a functor between the underlying categories which carries every covering family of C to a covering family of C'. The functor u is not continuous in general (cf. however Proposition 1.6). Here is a counterexample. Let S be an infinite set belonging to $\mathcal{U}$, and let C be the full subcategory of the category of sets whose objects are the finite nonempty subsets of S. Endow C with the canonical topology. Notice that the covering families of C are the surjective families of maps, and that covering families are nonempty. Notice also that, up to isomorphism, there are only two constant sheaves on C: the sheaf with value the empty set and the sheaf with value a one-point set. Put $C = C'$, and let $u : C \to C'$ be a constant functor. The functor u carries covering families of C to covering families of C', but u is not continuous. Indeed, since the representable presheaves of C' are sheaves, for every object X of C' there exists a sheaf F on C' such that the cardinality of $F(X)$ is at least 2; consequently the presheaf u^*F is not a sheaf.

2 Cocontinuous functors

Definition 2.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *cocontinuous* if it has the following property:

(COC) For every object Y of C, and every sieve $R \hookrightarrow u(Y)$ covering $u(Y)$, the sieve of Y generated by the arrows $Z \to Y$ such that $u(Z) \to u(Y)$ factors through R, covers Y.

Proposition 2.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. Let

$$\widehat{u}_* : \widehat{C} \to \widehat{C'}$$

denote the functor right adjoint to $F \mapsto F \circ u = \widehat{u}^* F$ (Exposé I, 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C, $\widehat{u}_* G$ is a sheaf on C'.

Proof. *Sufficiency.* Suppose that for every sheaf G on C, the presheaf $\widehat{u}_* G$ is a sheaf. Let Y be an object of C, and let $R \hookrightarrow u(Y)$ be a covering sieve. We have an isomorphism

$$\mathrm{Hom}(u(Y), u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G),$$

whence, by adjunction (Exposé I, 5.1), an isomorphism

$$\mathrm{Hom}(\widehat{u}^* u(Y), G) \xrightarrow{\sim} \mathrm{Hom}(\widehat{u}^* R, G).$$

It follows (Exposé II, 5.3) that the morphism

$$\widehat{u}^* R \to \widehat{u}^* u(Y)$$

is bicovering. Since $\widehat{u}^*$ commutes with projective limits, this morphism is a monomorphism, and therefore is a covering monomorphism. On the other hand, there is a canonical morphism

$$Y \xrightarrow{p} \widehat{u}^* u(Y)$$

deduced from the adjunction morphism $\mathrm{id}_C \to \widehat{u}^* u_!$ (Exposé I, 5.4(3)). By base change along p, we obtain a covering sieve of Y:

$$\widehat{u}^* R \times_{\widehat{u}^* u(Y)} Y \to Y.$$

The arrows $Z \to Y$ which factor through this sieve are precisely those described in property (COC).

Necessity. Let S be an object of C' and let $R \hookrightarrow S$ be a sieve covering S. We must prove that, for every sheaf G on C, the map

$$\mathrm{Hom}(S, u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G)$$

is an isomorphism. Using Proposition II, 5.3 and the adjunction isomorphisms, it is enough to prove that the monomorphism

$$\widehat{u}^* R \to \widehat{u}^* S$$

is covering. For this, it is enough to prove (Exposé II, 5.1) that, for every base change $Y \to \widehat{u}^* S$, where Y is an object of C, the sieve

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y$$

is covering. By the properties of adjoint functors and Exposé I, 5.4(3), the morphism $Y \to \widehat{u}^* S$ factors as

$$Y \xrightarrow{p} \widehat{u}^* u(Y) \to \widehat{u}^* S.$$

Thus the sieve $\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$ is obtained by base change along p from the monomorphism

$$\widehat{u}^* R \times_{\widehat{u}^* S} \widehat{u}^* u(Y) \hookrightarrow \widehat{u}^* u(Y).$$

Since $\widehat{u}^*$ commutes with projective limits, this monomorphism is the image under $\widehat{u}^*$ of the covering sieve

$$R \times_S u(Y) \hookrightarrow u(Y).$$

The hypothesis (COC) therefore implies that

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$$

is covering. $\square$

Proposition 2.3. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a cocontinuous functor. Let

$$u^* : \widetilde{C'} \to \widetilde{C}$$

be the functor $u^* = \underline{a}\widehat{u}^* i'$, where $\underline{a}$ denotes the associated-sheaf functor for $\widehat{C}$, $\widehat{u}^*$ the functor $F \mapsto F \circ u$, and i' the inclusion functor $\widetilde{C'} \to \widehat{C'}$.

1) The functor u^* commutes with inductive limits and is exact.

2) There is a canonical isomorphism

$$u^* \underline{a}' \simeq \underline{a}\widehat{u}^*,$$

where $\underline{a}'$ denotes the associated-sheaf functor for $\widehat{C'}$.

3) The functor u^* has a right adjoint $u_* : \widetilde{C} \to \widetilde{C'}$. When C is small, the diagram

$$\begin{tikzcd} \widetilde{C} \arrow[r, "u_*"] \arrow[d, hook, "i"'] & \widetilde{C'} \arrow[d, hook, "i'"] \\ \widehat{C} \arrow[r, "\widehat{u}_*"] & \widehat{C'} \end{tikzcd} \tag{2.3.1}$$

in which the lower functor is right adjoint to $\widehat{u}^*$ (Exposé I, 5.1), is commutative up to canonical isomorphism.

4) Let $\mathcal{V} \supset \mathcal{U}$ be a universe. Let

$$u_{*\mathcal{U}} : \widetilde{C}_{\mathcal{U}} \to \widetilde{C'}_{\mathcal{U}} \quad \text{(respectively } u_{*\mathcal{V}} : \widetilde{C}_{\mathcal{V}} \to \widetilde{C'}_{\mathcal{V}}\text{)}$$

denote the right adjoint relative to the universe $\mathcal{U}$ (respectively $\mathcal{V}$) whose existence is asserted in (3). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u_{*\mathcal{U}}"] \arrow[d, hook] & \widetilde{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u_{*\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{2.3.2}$$

commutes up to isomorphism.

Proof. Assertions (1) and (2) follow immediately from Exposé II, 3.4. Assertion (3), when C is small, follows from Proposition 2.2. When C is a $\mathcal{U}$-site, it will be proved in 4.4. Assertion (4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic, and from Exposé I, 3.5. When the topologies on C and C' are chaotic, the commutativity of (2.3.2) is immediate from the explicit description of u_* (Exposé I, 5.1). □

2.4

We shall not develop here the considerations concerning γ-objects of categories of sheaves. It will be enough for us to observe that the functors u^* and u_* introduced in this number always commute with finite projective limits, unlike what happened in the preceding number for the functor u^s. Consequently they extend naturally to functors defined on γ-objects; these extensions are adjoint to one another and commute with the "underlying sheaf of sets" functors. We shall make the abuses of notation indicated in 1.8, writing u^* and u_* for the extensions to γ-objects.

Proposition 2.5. Let C and C' be two $\mathcal{U}$-sites and let

$$C \underset{u}{\overset{v}{\rightleftarrows}} C'$$

be a pair of functors, where v is left adjoint to u. The following properties are equivalent:

i) The functor u is continuous.

ii) The functor v is cocontinuous.

Moreover, under these equivalent conditions, there are canonical isomorphisms $v_* \simeq u_s$ and $v^* \simeq u^s$. In particular, u^s commutes with finite projective limits.

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathcal{U}$ for which C and C' are $\mathcal{U}$-sites; this is immediate for cocontinuous functors, and follows from Proposition 1.5 for continuous functors. Thus, after enlarging the universe if necessary, we may suppose that C and C' are small. The proposition then follows from Proposition 2.2 and Exposé I, 5.6. □

Proposition 2.6. Let $u : C \to C'$ be a continuous and cocontinuous functor between two $\mathcal{U}$-sites. Then the functor

$$u^* : \widetilde{C'} \to \widetilde{C}$$

(Proposition 2.3) commutes with $\mathcal{U}$-inductive and projective limits. Let

$$u_! : \widetilde{C} \to \widetilde{C'}, \qquad u_* : \widetilde{C} \to \widetilde{C'}$$

denote the left and right adjoints of u^*, respectively. The functor $u_!$ is fully faithful if and only if u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful; the converse holds when the topologies of C and C' are less fine than the canonical topology.

The functor u^* has a right adjoint u_* (Proposition 2.3) and a left adjoint $u_!$ (Proposition 1.2). Hence u^* commutes with inductive and projective limits (Exposé I, 2.11). The fact that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (Exposé I, 5.7(1)). When u is fully faithful, the functor

$$\widehat{u}_* : \widehat{C}_{\mathcal{V}} \to \widehat{C'}_{\mathcal{V}},$$

relative to $\mathcal{V}$-presheaves for a sufficiently large universe $\mathcal{V}$, is fully faithful (Exposé I, 5.7). Therefore $u_* : \widetilde{C} \to \widetilde{C'}$ is fully faithful by the commutativity of (2.3.1) and (2.3.2), and hence $u_!$ is fully faithful by what precedes. The converse follows from the existence of the commutative diagram in Proposition 1.2(iv), taking into account the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are less fine than the canonical topology. □

Example 2.7. With the notation of 1.9.2, the functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X)$$

is continuous (1.9.2) and cocontinuous (Proposition 2.2). Moreover, let $i : Y \to X$ be the canonical injection and let $i^{-1} : \mathrm{Open}(X) \to \mathrm{Open}(Y)$ be the functor it defines (1.9.1). The functor i^{-1} is right adjoint to Φ. Thus there is a sequence of three adjoint functors

$$\Phi^s, \quad \Phi_s = \Phi^*, \quad \Phi_*,$$

and canonical isomorphisms

$$\Phi^* \simeq (i^{-1})^s, \qquad \Phi_* \simeq i_s^{-1}.$$

3 Induced topology

3.1

Let C' be a site, let C be a category, and let $u : C \to C'$ be a functor. For every universe $\mathcal{U}$ such that C' is a $\mathcal{U}$-site and C is a $\mathcal{U}$-small category, denote by $\mathcal{T}_{\mathcal{U}}$ the finest among the topologies T on C which make u continuous (Definition 1.1). Such a topology exists by Exposé II, 2.2. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on $\mathcal{U}$. Indeed, if $\mathcal{V} \supset \mathcal{U}$ is a universe, one has $\mathcal{T}_{\mathcal{U}} = \mathcal{T}_{\mathcal{V}}$ (Definition 1.1 and Proposition 1.5). The topology $\mathcal{T}_{\mathcal{U}}$ is called the *topology induced on C by the topology of C' by means of the functor u.*[5]

Proposition 3.2. Let C be a small category, let C' be a $\mathcal{U}$-site, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by u. Let X be an object of C and let $R \to X$ be a sieve on X. The following properties are equivalent:

i) The sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$.

[5]When no confusion can result, this topology is called simply the *topology induced on C by the topology of C'*.

ii) For every base change $Y \to X$, where Y is an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2).

Proof. i) $\Rightarrow$ ii) follows from axiom (T1) for topologies and from Proposition 1.2.

ii) $\Rightarrow$ i). For every sheaf F on C', the map

$$\mathrm{Hom}(Y, u^*F) \to \mathrm{Hom}(R \times_X Y, u^*F)$$

is isomorphic, by adjunction (Exposé I, 5.1), to the map

$$\mathrm{Hom}(u(Y), F) \to \mathrm{Hom}(u_!(R \times_X Y), F),$$

which is bijective (Exposé II, 5.3). Hence, by Exposé II, 2.2, the sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.3. Let C' be a site, let C be a category, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by the topology of C'. Let $(X_i \to X)_{i \in I}$ be a family of base-changeable morphisms of C, and suppose that u commutes with fiber products.[6] The following properties are equivalent:

i) The family $(X_i \to X)_{i \in I}$ is covering for $\mathcal{T}$.

ii) The family $(u(X_i) \to u(X))_{i \in I}$ is covering.

Proof. i) $\Rightarrow$ ii) follows from Proposition 1.6.

ii) $\Rightarrow$ i). Let $\mathcal{U}$ be a universe such that C is $\mathcal{U}$-small and C' is a $\mathcal{U}$-site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \to X$. The presheaf R is the cokernel of the pair of arrows (Exposé I, 2.12 and 3.5)

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i,$$

where the coproduct is taken in $\widehat{C}$. Since $u_!$ commutes with inductive limits (Exposé I, 5.4), the presheaf $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

Since u commutes with fiber products, $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i),$$

and hence $u_!R \to u(X)$ is the sieve on $u(X)$ generated by the $u(X_i) \to u(X)$, $i \in I$. Using once again that u commutes with fiber products, one shows that, for every base change $Y \to X$ with Y an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

[6] In fact, it is enough that u commute with fiber products of the form $X_i \times_X X'$.

is the sieve on $u(Y)$ generated by the morphisms

$$u(X_i) \times_{u(X)} u(Y) \to u(Y), \qquad i \in I.$$

Since $(u(X_i) \to u(X))_{i\in I}$ is covering, the family $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$, is covering. Thus $u_!(R \times_X Y) \to u(Y)$ is a covering sieve, and therefore by Proposition 3.2 the sieve $R \to X$ is covering for $\mathcal{T}$. ☐

Corollary 3.4. Let C' be a $\mathcal{U}$-site, let C be a full subcategory of C', and let $u : C \to C'$ be the inclusion functor. Suppose that fiber products are representable in C and that u commutes with fiber products. The following conditions are equivalent:

- i) a) For every object X of C, every covering family $(Y_j \to X)_{j\in J}$ of C' is dominated by a covering family $(X_i \to X)_{i\in I}$, where the X_i are objects of C.
  - b) There exists a small set G of objects of C such that every object of C is the target of a covering family in C' of morphisms whose sources belong to G.
- ii) The topology induced by the topology of C' is a $\mathcal{U}$-topology (Exposé I, 3.0.2), and u is continuous and cocontinuous for this topology.

Proof. This follows immediately from Corollary 3.3 and Definition 2.1. ☐

Proposition 3.5. Let C be a $\mathcal{U}$-site and let $\epsilon_C : C \to \widetilde{C}$ be the canonical functor (Exposé II, 4.4.0). Endow $\widetilde{C}$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $\widetilde{C}$.

Proof. The covering families of $\widetilde{C}$ for the canonical topology are the universal effective epimorphic families (Exposé II, 2.5), that is (Exposé II, 4.3), the epimorphic families. Let T be the topology of the site C, and let $\mathcal{T}_{\mathcal{U}}$ be the finest topology T' on C such that, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf for T'. It is enough to show that $T = \mathcal{T}_{\mathcal{U}}$; then $\mathcal{T}_{\mathcal{U}}$ is a $\mathcal{U}$-topology, and hence $\mathcal{T}_{\mathcal{U}}$ is the induced topology.

A) The topology T is finer than $\mathcal{T}_{\mathcal{U}}$. Let $(X_i \to X)_{i\in I}$ be a covering family for $\mathcal{T}_{\mathcal{U}}$. Then for every sheaf F on $\widetilde{C}$, the map

$$F(\epsilon_C(X)) \to \prod_i F(\epsilon_C(X_i))$$

is injective. Hence (Exposé II, 5.2) the family $(\epsilon_C X_i \to \epsilon_C X)_{i\in I}$ is covering for the canonical topology of $\widetilde{C}$. Therefore (Exposé II, 5.2) $(X_i \to X)_{i\in I}$ is covering for T.

B) The topology T is less fine than $\mathcal{T}_{\mathcal{U}}$. It is enough to show that $\epsilon_C : C \to \widetilde{C}$ is continuous (Definition 1.1). But it will be proved in Exposé IV, no. 1, that every sheaf on $\widetilde{C}$ is representable; hence, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf on C. We shall not use Proposition 3.5 before Exposé IV, 1.

We record two results which will be used in Exposé VI, 7. ☐

Proposition 3.6. Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathcal{T}_{\mathcal{U}}$ on C which is the least fine among the topologies for which the u_i are continuous. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

The last assertion of Proposition 3.6 follows from Proposition 1.5.

Let T be a topology on C. The functors u_i are continuous if and only if, for every $i \in I$ and every covering sieve $R \hookrightarrow X$ of an object X of C_i, the morphism

$$u_{i!}(R) \to u_i(X)$$

of $\widehat{C}$ is bicovering (Proposition 1.2(ii)). Thus Proposition 3.6 is a consequence of the following lemma.

Lemma 3.6.1. Let C be a small category and let $(u_i : F_i \to G_i)_{i \in I}$ be a family of arrows of $\widehat{C}$. Then there exists on C a least fine topology among those which make the morphisms u_i covering (respectively bicovering) (Exposé II, 5.2).

To say that $u : F \to G$ is covering for a given topology T means that, for every arrow $X \to G$, with $X \in \operatorname{Ob} C$, the corresponding arrow $F \times_G X \to X$ is covering; equivalently, the family of arrows $X' \to X$ of C which factor through that arrow is covering. The existence of a least fine topology among those for which the $u_i : F_i \to G_i$ are covering follows from Exposé I, 1.1.6, proving the non-parenthesized assertion of Lemma 3.6.1. The parenthesized assertion follows by recalling that a morphism $u : F \to G$ is bicovering if and only if the morphisms $u : F \to G$ and $\Delta_u : F \to F \times_G F$ are covering.

Proposition 3.7. Let $(C_i)_{i \in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathfrak{T}_{\mathcal{U}}$ which is the finest topology for which the u_i are cocontinuous. The topology $\mathfrak{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

Let $\mathcal{U}$ be a universe for which the categories considered are small. The functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i, the presheaf $\widehat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (Proposition 2.2). It follows that $\mathfrak{T}_{\mathcal{U}}$ is the finest topology for which the presheaves $\widehat{u}_{i*}F$, $i \in I$, $F \in \mathrm{Ob}(C_i)$, are sheaves (Exposé II, 2.2). The last assertion follows from Proposition 2.2.

4 The comparison lemma

Theorem 4.1 (comparison lemma). Let C be a small category, let C' be a site whose underlying category is a $\mathcal{U}$-category, and let $u : C \to C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the properties:

i) Every object of C' can be covered by objects coming from C.

ii) The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C.

One always has i) $\Rightarrow$ ii). When C' is a $\mathcal{U}$-site and the topology of C' is less fine than the canonical topology, one has ii) $\Rightarrow$ i).

4.1.1

We first prove i) $\Rightarrow$ ii). The proof is in two steps.

First step. For every presheaf H of $\widehat{C'}$, the adjunction morphism

$$\varphi : u_! u^* H \to H$$

(Exposé I, 5.1) is bicovering (Exposé II, 5.3), and the functor $u_s : \widetilde{C'} \to \widetilde{C}$ (1.1.1) is fully faithful.

Let C/H be the small category whose objects are pairs consisting of an object Y of C and a morphism $u(Y) \to H$, and whose morphisms are the commutative diagrams

$$\begin{tikzcd} u(Y) \arrow[rr, "u(m)"] \arrow[dr] & & u(Y') \arrow[dl] \\ & H & \end{tikzcd} .$$

We have

$$u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} Y$$

(Exposé I, 3.4), and hence (Exposé I, 5.4)

$$u_! u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} u(Y).$$

The adjunction morphism is the evident morphism coming from this description of $u_! u^* H$ as an inductive limit. It has the following property:

($*$) For every object Y of C, every morphism $m : u(Y) \to H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \to u_! u^* H$ followed by φ.

It follows immediately from (i) that φ is covering (Exposé II, 5.1). Let us prove that it is bicovering. Let $p, q : Z \rightrightarrows u_! u^* H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \to Z$, one has $\varphi p n = \varphi q n$. Property ($*$) then implies $pn = qn$, and since the $u(Y)$ cover Z, the kernel of (p, q) is a covering sieve of Z. Thus φ is bicovering (Exposé II, 5.3). For every sheaf H on C', $u^* H$ is a sheaf on C, denoted $u_s H$ (1.1.1). Moreover

$$u^s u_s H = \underline{a}' u_! u_s H$$

(Proposition 1.3), and the adjunction morphism $u^s u_s H \to H$ is obtained by applying the associated-sheaf functor to $\varphi : u_! u^* H \to H$. Therefore (Exposé II, 5.3) the adjunction morphism $u^s u_s H \to H$ is an isomorphism. Hence $u_s : \widetilde{C'} \to \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous and the functor $u^s : \widetilde{C} \to \widetilde{C'}$ (Proposition 1.2) is fully faithful. Therefore $u_s : \widetilde{C'} \to \widetilde{C}$ is an equivalence.

Let Y be an object of C, and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since u^* commutes with projective limits (Exposé I, 5.5), the morphism

$$u^*(i) : u^*(R) \hookrightarrow u^* u(Y)$$

is a monomorphism. Since u is fully faithful, $u^*u(Y) \simeq Y$, and hence this gives a sieve $u^*(R) \to Y$ on Y. To show that u is cocontinuous, it is enough to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is a covering sieve for the induced topology on C (Definition 2.1). For this it is enough to prove (Proposition 3.2) that, for every base change $m : X \to Y$, the morphism

$$u_!(u^*(R) \times_Y X) \to u(X)$$

is bicovering. Since u^* commutes with projective limits, one has

$$u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X)),$$

and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It is therefore enough to prove that, for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $\widehat{C'}$

$$u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$$

is bicovering. But this morphism factors as the adjunction morphism $u_!u^*(R) \to R$, which is bicovering by the first step, followed by the monomorphism $R \hookrightarrow u(Y)$, which is covering. Hence it is bicovering (Exposé II, 5.3). This proves that u is cocontinuous. Since u is continuous and fully faithful, Proposition 2.6 (used in the case where C is small) implies that u^s (denoted $u_!$ in Proposition 2.6) is fully faithful. Since u^s and u_s are adjoint to one another and both are fully faithful, they are quasi-inverse functors; hence u_s is an equivalence.

4.1.2

We now prove ii) $\Rightarrow$ i). The objects Y of C form a generating family of $\widetilde{C}$ (Exposé II, 4.10). Therefore, for every object X of C', there exists an epimorphic family in $\widetilde{C}$

$$v_i : Y_i \to u_s X, \qquad i \in I.$$

We have $u_s u(Y_i) = Y_i$, and, since u_s is an equivalence of categories, $v_i = u_s(w_i)$, where the $w_i : u(Y_i) \to X$ form an epimorphic family of $\widetilde{C'}$. It follows from Exposé II, 5.1, that the family $w_i : u(Y_i) \to X$, $i \in I$, is covering.

4.1 End of the proof of Proposition 1.2

Let G be a full subcategory of C whose objects form a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$ (3.1). The functor $(u \circ i)_s = i_s \circ u_s$ (1.1.1) has a left adjoint by the first part of the proof of Proposition 1.2. Since i_s is an equivalence of categories (Theorem 4.1), the functor u_s has a left adjoint u^s, and there is a functorial isomorphism

$$u^s \simeq (u \circ i)^s \circ i_s.$$

Let us prove that the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd} \tag{4.2.1}$$

commutes up to isomorphism. For every sheaf H on C' and every $X \in \operatorname{Ob} C$,

$$\operatorname{Hom}_{\widetilde{C'}}(\epsilon_{C'} uX, H) \simeq u_s H(X).$$

Moreover,

$$\operatorname{Hom}_{\widetilde{C'}}(u^s \epsilon_C X, H) \simeq \operatorname{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H) \simeq u_s H(X),$$

by adjunction and by the definition of ϵ_C. Hence $\epsilon_{C'} uX \simeq u^s \epsilon_C X$ for every $X \in \operatorname{Ob} C$. This proves i) $\Rightarrow$ iv). Let us prove iv) $\Rightarrow$ i). There is a commutative diagram

$$\begin{tikzcd} G \arrow[r, "i"] \arrow[d, "\epsilon_G"'] & C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"'] \\ \widetilde{G} \arrow[r, "i^s"] & \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}, \end{tikzcd} \tag{4.2.2}$$

and therefore, by the first part of the proof, the functor $u \circ i : G \to C'$ is continuous. From Theorem 4.1 and Definition 1.1 it follows immediately that u is continuous, hence iv) $\Rightarrow$ i). One also obtains the uniqueness of u^s, knowing from the first part of the proof the uniqueness when C is small.

4.2 End of the proof of Propositions 1.5 and 1.7

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$. It follows immediately from Theorem 4.1 and Definition 1.1 that u is continuous if and only if $u \circ i$ is continuous. From this observation and from the first part of the proof of Proposition 1.5 the general case follows. This observation also reduces the proof of Proposition 1.7 to the case where C is small.

4.3 End of the proof of Proposition 2.3

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1), endowed with the topology induced by the inclusion functor $i : G \to C$. It follows from the proof of Theorem 4.1 (second step in 4.1.1) that i is cocontinuous. Therefore, when u is cocontinuous, the functor $u \circ i : G \to C'$ is cocontinuous. Moreover,

$$i^* : \widetilde{C} \to \widetilde{G}$$

is an equivalence of categories (Theorem 4.1). Thus the functor $u^* : \widetilde{C'} \to \widetilde{C}$ has a right adjoint. This proves assertion (3). To prove assertion (4), it is enough to observe that

$$(u \circ i)_* \simeq u_* \circ i_*$$

and that i_* is an equivalence (Theorem 4.1). The commutativity of (2.3.2) follows from the commutativity of these diagrams when C is small.

5 Localization

5.1

Let now C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. Unless explicitly stated otherwise, the category C/X will be endowed with the topology $\mathcal{T}$ induced by the functor

$$j_X : C/X \to C$$

(3.1). The notation C/X will denote the category C/X endowed with the topology $\mathcal{T}$. Proposition I, 5.11 shows that the functor $j_{X!}$ commutes with fiber products and therefore carries every monomorphism to a monomorphism. In particular, for every object $(Y \to X)$ of C/X, the functor $j_{X!}$ gives a bijective correspondence between the sieves, in the category C/X, of the object $(Y \to X)$, and the sieves, in C, of the object Y.

Proposition 5.2. Let C be a $\mathcal{U}$-site, let X be an object of $\widehat{C}$, and let $j_X : C/X \to C$ be the corresponding continuous functor.

1) A sieve R of an object $(Z \to X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C.

2) The functor $j_X : C/X \to C$ is cocontinuous and continuous.

3) Let $m : Y \to X$ be a morphism of $\widehat{C}$. Then there is a commutative diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd}.$$

The topology induced by j_Y on C/Y is equal to the topology induced by j_m on C/Y.

4) The topology induced by $j_X : C/X \to C$ is a $\mathcal{U}$-topology (Exposé II, 3.0.2).

Proof.

1) If the sieve R of $(Z \to X)$ is covering, the sieve $j_{X!}(R) \hookrightarrow Z$ is covering (Proposition 1.6). Conversely, if $j_{X!}(R) \to Z$ is covering in C, the same is true for every sieve obtained by base change in C/X. Hence R is covering (Proposition 3.2).

2) This follows immediately from (1) by applying Definition 2.1.

3) This follows immediately from the description of covering sieves given in (1).

4) Let $(G_i)_{i \in I}$ be a small topologically generating family of C. One checks immediately that the small family of pairs

$$(u : G_i \to X), \qquad u \in \coprod_{i \in I} \mathrm{Hom}_{\widehat{C}}(G_i, X),$$

is topologically generating in C/X.

$\square$

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors (4.3.2) between categories of sheaves of sets (Propositions 1.3 and 2.3):

$$j_X^s : (C/X)^\sim \to \widetilde{C}, \qquad j_X^* = j_{X,s} : \widetilde{C} \to (C/X)^\sim, \qquad j_{X*} : (C/X)^\sim \to \widetilde{C}.$$

In the special situation of Proposition 5.2, we shall use the following terminology and notation:

1) The functor j_{X*} will be called the *direct image functor*.

2) The functor $j_{X,s} = j_X^*$ will be denoted j_X^* and called the *restriction functor to C/X*.

3) The functor j_X^s on sheaves of sets will be called the "extension by the empty set to the category C" functor and will be denoted $j_{X!}$ (cf. 2.9.2).

Thus there is a sequence of three adjoint functors between $(C/X)^\sim$ and $\widetilde{C}$:

$$j_{X!}, \quad j_X^*, \quad j_{X*}.$$

Proposition 5.4. The functor $j_{X!} : (C/X)^\sim \to \widetilde{C}$ factors through the category $\widetilde{C}/\underline{a}X$, where $\underline{a}$ is the associated-sheaf functor:

$$(C/X)^\sim \xrightarrow{e_X^\sim} \widetilde{C}/\underline{a}X \to \widetilde{C}.$$

The functor

$$e_X^\sim : (C/X)^\sim \to \widetilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction functor to C/X, composed with the equivalence $e_X^\sim$, is isomorphic to the "base change by $\underline{a}X \to \epsilon$" functor

$$F \longmapsto (F \times \underline{a}X \xrightarrow{\mathrm{pr}_2} \underline{a}X),$$

where ϵ is the final object of $\widetilde{C}$.

Proof. The image under $j_{X!}$ of the final object of $(C/X)^\sim$ is the object $\underline{a}X$, whence the factorization. To show that $e_X^\sim$ is an equivalence, we restrict ourselves to a few indications. By Exposé I, 5.11, a presheaf on C/X is defined by a presheaf F on C endowed with a morphism $F \to X$. One then proves that the following two properties are equivalent:

i) The presheaf on C/X defined by $F \to X$ is a sheaf.

ii) The following diagram is cartesian, where i and $\underline{a}$ denote the inclusion into presheaves and the associated-sheaf functor:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & i\underline{a}F \arrow[d] \\ X \arrow[r] & i\underline{a}X. \end{tikzcd}$$

It follows immediately that $e^{\sim}_X$ is an equivalence. The last assertion is trivial. $\square$

Proposition 5.5.

1) Let C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. The following diagram of categories and functors is commutative up to canonical isomorphisms:

$$\begin{tikzcd}
C/X \arrow[r, "h_X"] \arrow[d] & \widehat{(C/X)} \arrow[r, "\underline{a}_X"] \arrow[d, "\widehat{e}_X"] & (C/X)^{\sim} \arrow[rr, "i_X"] \arrow[d, "e^{\sim}_X"] & & \widehat{(C/X)} \arrow[d, "\widehat{e}_X/X"] \\
C/X \arrow[r, "h/X"] & \widehat{C}/X \arrow[r, "\underline{a}/X"] & \widetilde{C}/\underline{a}X \arrow[r, "i/\underline{a}X"] & \widehat{C}/i\underline{a}X \arrow[r, "\pi"] & \widehat{C}/X.
\end{tikzcd}$$

The notation $\underline{a}_X$ and i_X denotes the associated-sheaf functor and the inclusion into presheaves for the site C/X. The notation $\underline{a}/X$ denotes the natural extension of $\underline{a}$, the associated-sheaf functor for the site C, to the category of arrows with target X; similarly for $i/\underline{a}X$. Finally, π denotes the base-change functor by the canonical arrow $X \to i\underline{a}X$.

2) Suppose in addition that $m : Y \to X$ is a morphism of $\widehat{C}$. The following diagram is commutative up to canonical isomorphism:

$$\begin{tikzcd}
\widetilde{C}/\underline{a}X/\underline{a}Y \arrow[d, equal] & (C/X)^{\sim}/\underline{a}_X Y \arrow[l, "f"'] & \\
& & (C/X/Y)^{\sim} \arrow[ul, "g"'] \arrow[dl, "e^{\sim}_m"] \\
\widetilde{C}/\underline{a}Y & (C/Y)^{\sim} \arrow[l, "e^{\sim}_Y"'] & .
\end{tikzcd}$$

The arrow f is the natural extension of the equivalence $e^{\sim}_X$ to the category of arrows with target $\underline{a}_X Y$, and therefore is an equivalence of categories. The arrow g is none other than the equivalence of Proposition 5.4 applied to the situation $(C/X)/[m]$.

3) Let

$$j_{\underline{a}X!} : \widetilde{C}/\underline{a}X \to \widetilde{C}$$

denote the functor "forget $\underline{a}X$", and let

$$j^*_{\underline{a}X} : \widetilde{C} \to \widetilde{C}/\underline{a}X$$

denote the functor "product with $\underline{a}X$". The following diagrams are commutative up to canonical isomorphism:

$$\begin{tikzcd}
(C/X)^{\sim} \arrow[r, "j_{X!}"] \arrow[d] & \widetilde{C} \arrow[d, equal] & \widetilde{C} \arrow[r, "j^*_X"] \arrow[d, equal] & (C/X)^{\sim} \arrow[d, "e^{\sim}_X"] \\
\widetilde{C}/\underline{a}X \arrow[r, "j_{\underline{a}X!}"] & \widetilde{C} & \widetilde{C} \arrow[r, "j^*_{\underline{a}X}"] & \widetilde{C}/\underline{a}X.
\end{tikzcd}$$

Proof. The proof of these assertions is immediate from the definition of the equivalences $e_X^{\sim}$ (Proposition 5.4) and $\widehat{e}_X$ (Exposé I, 5.11). □

References

[1] R. Godement, *Théorie des faisceaux*, Hermann, 1958, Act. Scient. Ind. no. 1252, Paris.